

WEEK 1 PROGRESS REPORT: PROJECT SETUP & LITERATURE REVIEW

1. GOALS & ACTIVITIES

- Finalized the research topic: Post-Disaster Assessment using Machine Learning and Semantic Communication.
- Studied related academic papers on post disaster management.

2. ADVISOR MEETING & ADVICE

- Meeting Date: 30/04/2026
- Advisor/Supervisor Name: Dr. Himal Suraweera
- Key Advice Received:
 - Advisor clarified about the goals and objectives of our project
 - Advised us to study the papers which are relevant to our project

We have already discussed with our advisor about the goals in semester 5. We have collected our own data as images from the places which were affected during Ditwa 2025 in SriLanka.

We have done 4 days of field work to collect our data from various places near Kandy.



POST DISASTER DAMAGE ASSESSMENT

The main goal of our project is to develop an intelligent system that can automatically assess disaster damage using advanced Machine Learning techniques.

The main purpose of our project is to assist:

- First responders in quickly identifying high-risk areas
- Local communities in understanding risk levels

In our system, we focus on four key AI techniques:

1.Classification – to identify the type of disaster or type of damage (for example, landslide, fallen tree, structural collapse).

2.Object Detection – to locate specific damaged structures within an image.

3.Segmentation – to precisely outline the affected areas, such as marking the exact region of a landslide.

4.Visual Question Answering (VQA) – to allow users to ask questions about the visual data.

Methodology & Workflow

Database Creation

to achieve our objective, we structured our work into clear stages. First, we focused on Database Creation. Currently, our main priority is building a robust and realistic disaster dataset. Since machine learning models depend heavily on data quality, we are carefully collecting and organizing real-world disaster images.

Validation.

We ensure that the collected data is suitable for training accurate models. This includes checking image clarity, angle, lighting conditions, and relevance to actual damage assessment.

User Integration.

We developed a mobile application that allows users to upload disaster images.

Field Work & Data Collection

We have already conducted extensive fieldwork to build our dataset. Our data collection was carried out on:

- January 10th, 2026
- January 11th, 2026
- January 18th, 2026
- January 19th, 2026

We visited multiple locations across the Central Province, including:

- Galaha Road
- Weligalla
- Gelioya
- Penideniya

During these visits, we captured real-world disaster evidence such as:

- Landslides
- Fallen trees
- Structural debris
- Damaged roads

OBJECTIVES

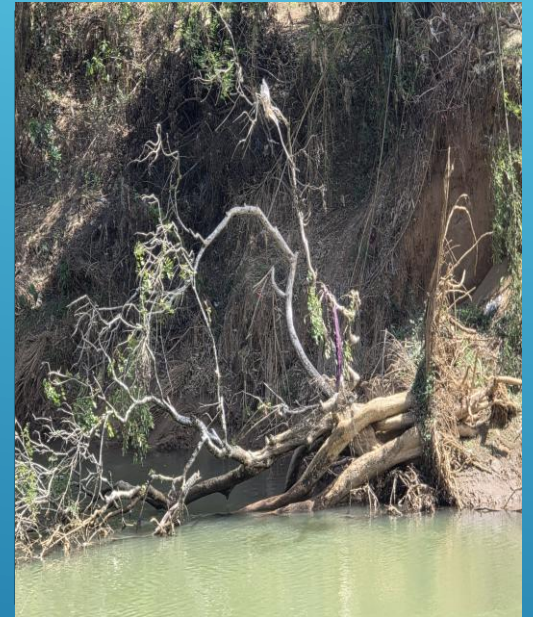
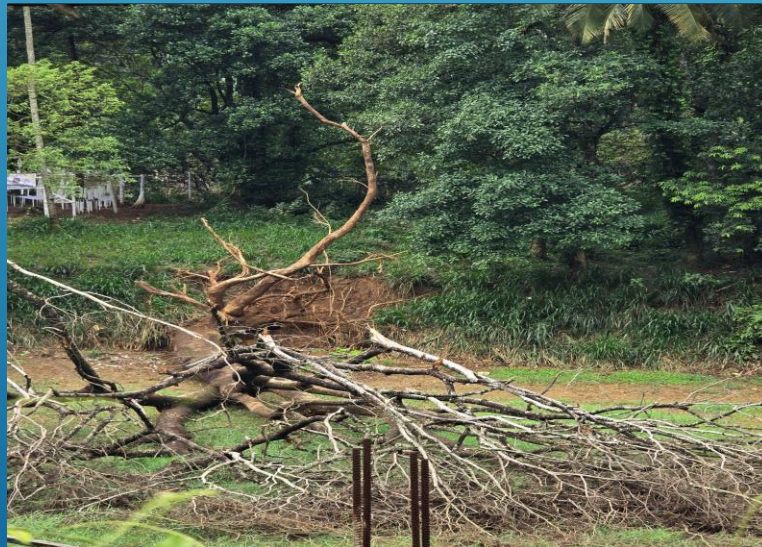
1.Achieving tasks using machine learning

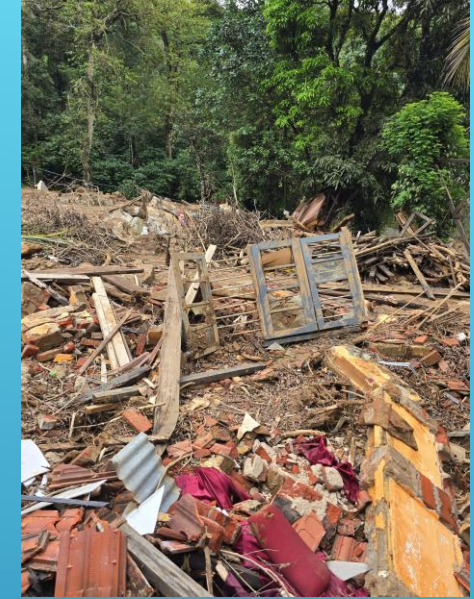
- Classification
- Object detection
- Segmentation
- VQA (Visual questioning & answering)

2.STEPS

- i. Create database.
- ii. Check whether they can be used for damage assessment.
- iii. Develop mobile apps to collect photos from users.
 - Assist users about how the photos should be taken.
 - Get data from audio data & text data.
- iv. Using drone images to make better.
- v. Using AI transmit only required data – semantic communication.

TREE UPROOTEN



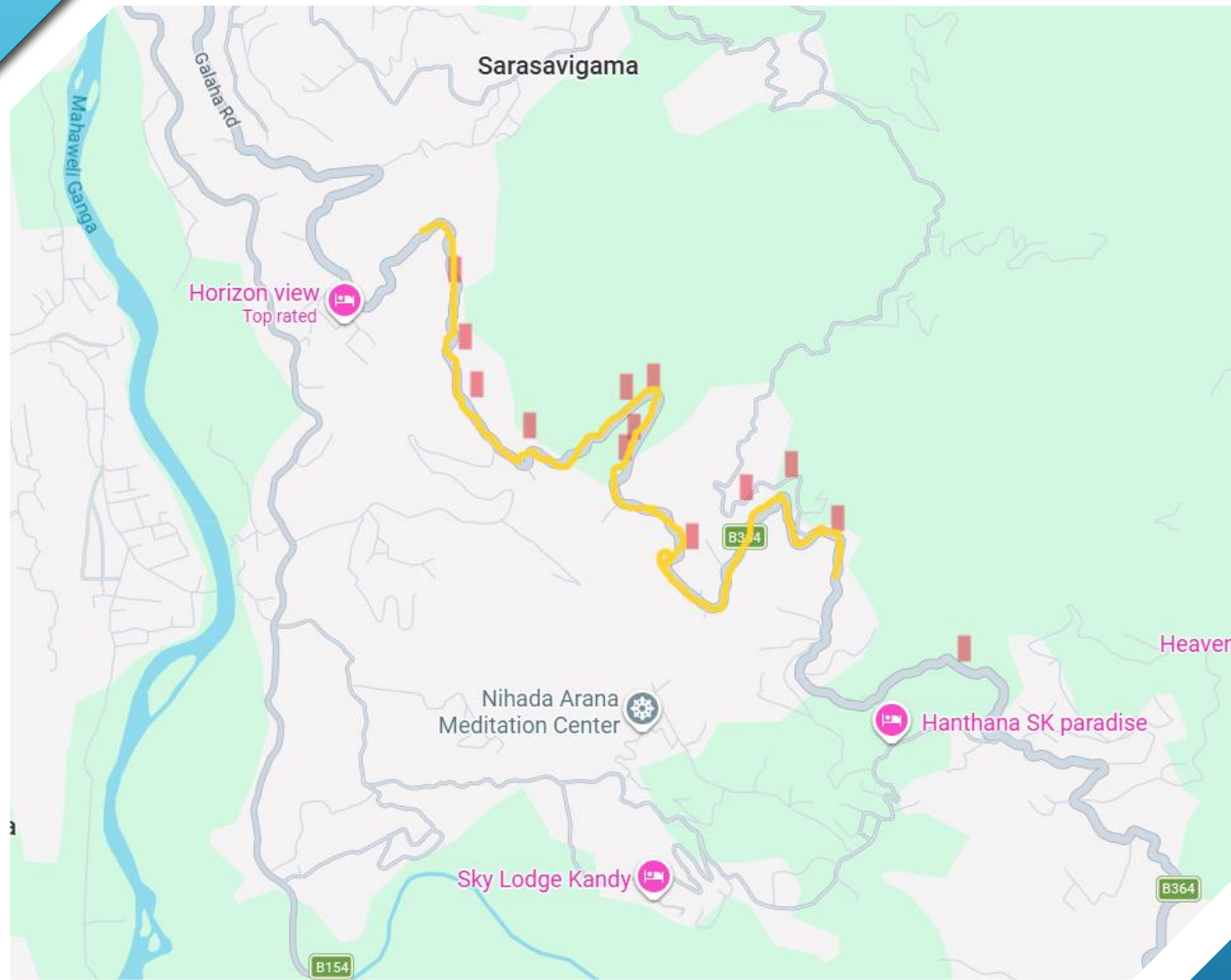


STRUCTURAL DEBRIS

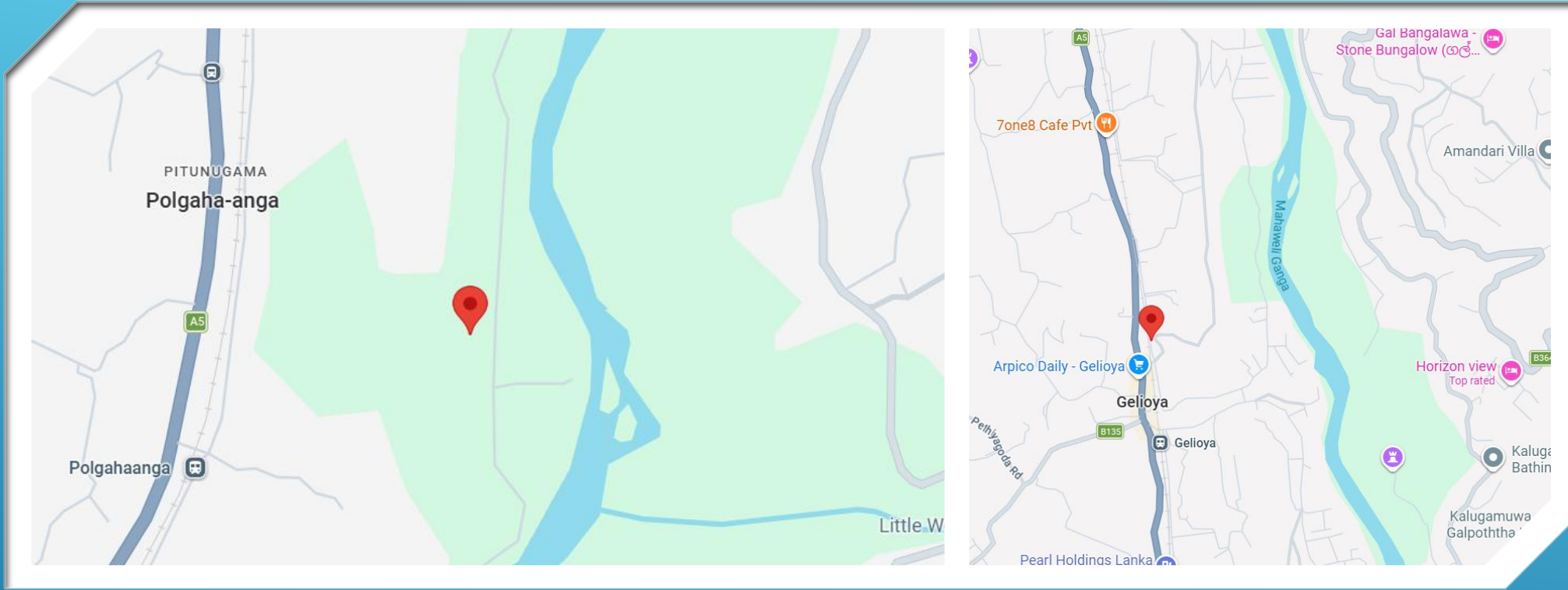




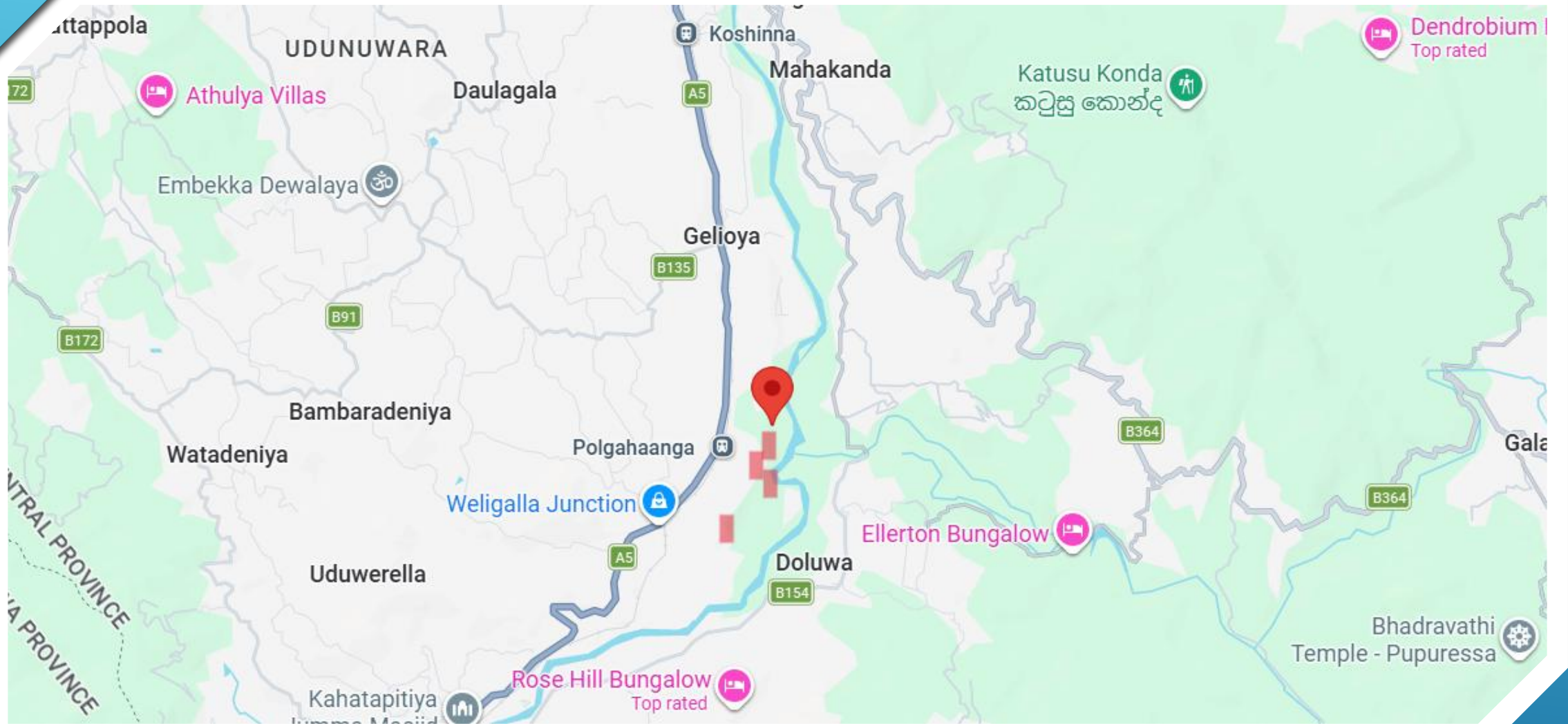
PLACES WHERE PICTURES WERE TAKEN ON 10/01/2026

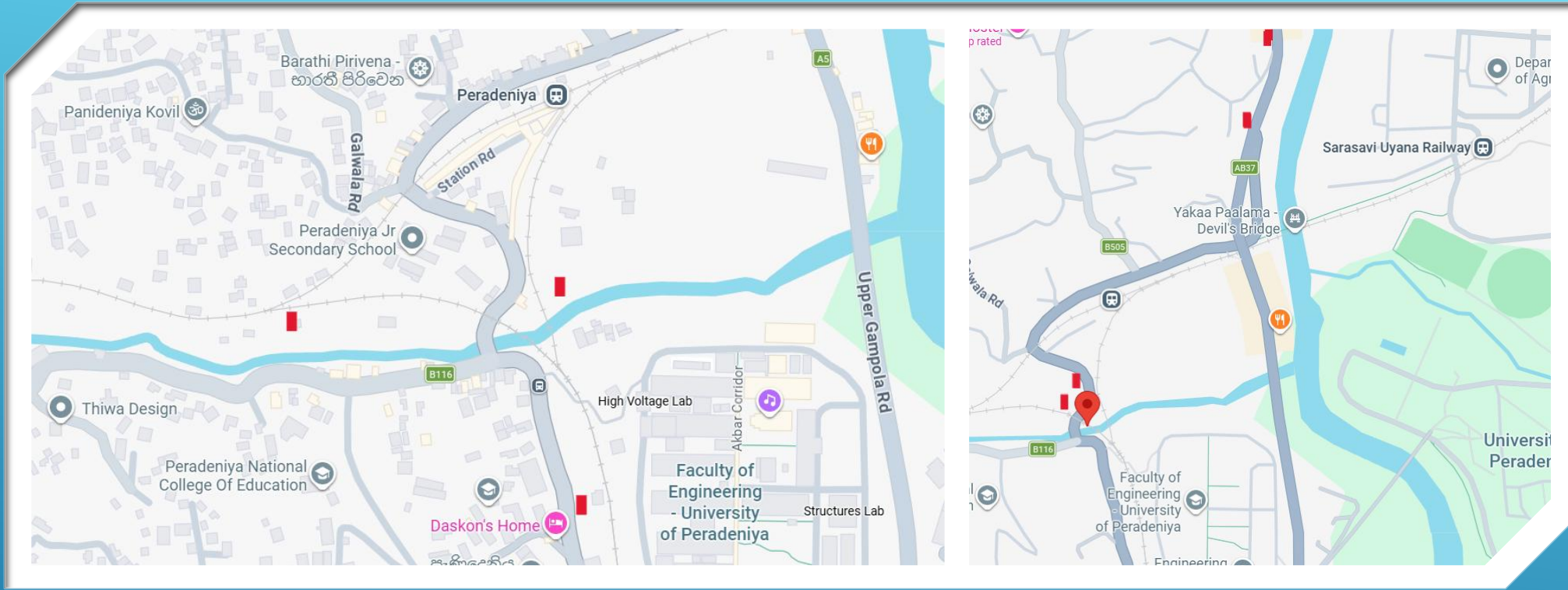






PLACES WHERE PICTURES WERE TAKEN
ON 11/01/2026





PLACES WHERE PICTURES WERE TAKEN
ON 18/01/2026 & 19/01/2026

Team Roles

1. App Development & System Infrastructure

This includes:

- Developing the mobile application that allows users to upload real-time disaster images.
- Designing the interface to guide users in capturing high-quality, usable photos.

2. Dataset Collection & Fieldwork

- Our dataset is built entirely from real-world data collected during field visits conducted on: January 10th, 11th, 18th, and 19th, 2026
- During these visits, we captured various types of disaster-related damage including landslides, fallen trees, road damage, and structural debris.
- Structuring and labeling the dataset correctly.
- Study existing disaster datasets.
- Identify common structures:
 - ✓ Image labels (damage level)
 - ✓ Bounding boxes (object detection)
 - ✓ Segmentation masks

3. Research Alignment

- Analyze **techniques used in research**:
 - ✓ Classification
 - ✓ Object Detection
 - ✓ Segmentation
 - ✓ VQA
- Compare those methods with our project
- Ensure our dataset follows similar standards and formats
- Help prepare data properly for model training

Complete Explanation: “A Machine Learning Approach for Post-Disaster Data Curation”

This document explains the system described in the paper in a clear, practical way, including examples and answers to questions such as what, why, and how each step works.

1. Problem the Paper Is Solving

After natural disasters (hurricanes, floods, earthquakes, etc.), thousands or millions of photos are collected from sources like:

- drones
- street cameras
- satellites
- rescue teams
- social media

Examples of such images may show:

- damaged houses
- broken roofs
- flooded streets
- debris around buildings

The main problem is that these images are usually stored randomly without proper labeling.

Because of this, researchers face a Data-Rich but Information-Poor situation (DRIP).

Example problem:

A researcher wants images of:

“Residential houses with roof damage caused by a hurricane.”

But the database may contain 10,000 or more images, and finding the correct ones manually can take hours or days.

Therefore, the paper proposes a system that uses Machine Learning and Natural Language Processing to automatically organize disaster images.

2. Overall Workflow of the System

The system processes each image using the following steps:

Image



Cloud Storage



Computer Vision detects objects



Remove faces / text for privacy



Generate secondary labels



NLP model calculates similarity



Primary labels are assigned



Store results in JSON database

Each step is explained below.

3. Step 1 – Image Input

The process starts when an image is uploaded into the system.

Examples of images:

- a hurricane-damaged house
- a flooded street
- a building with broken windows
- debris scattered around buildings

Example image contents:

- building
- broken roof
- flood water
- debris

The system receives this image as the input data.

4. Step 2 – Cloud Storage

After the image is uploaded, it is stored in cloud storage.

Examples of cloud platforms:

- AWS S3
- Google Cloud Storage
- Microsoft Azure Storage

Why cloud storage is used

Cloud storage allows the system to:

- store large numbers of images
- access them from anywhere
- process them automatically
- scale easily when more data arrives

Example flow:

User uploads an image

→ The image is stored in cloud storage

→ The processing system retrieves it for analysis.

5. Step 3 – Computer Vision Detects Objects

Next, the system uses Computer Vision models to analyze the image.

Computer Vision is an AI technique that allows computers to recognize objects inside images.

Examples of tools used:

- AWS Rekognition
- Google Vision API
- YOLO
- Mask R-CNN

The AI analyzes the image and produces object labels with confidence scores.

Example:

An image of a damaged house might produce the following labels:

building – 95%

flood – 82%

debris – 78%

tree – 70%

This means the AI believes the image contains those objects.

These labels are called secondary labels.

6. Step 4 – Removing Faces or Text for Privacy

Some disaster images may contain sensitive information such as:

- people's faces
- vehicle number plates
- shop signs
- street names

For privacy protection, the system automatically detects:

- faces
- text

Then it applies Gaussian blur to hide them.

Example:

Face detected → face is blurred

Text detected → text is blurred

This step ensures that personal information is not exposed when the images are stored or shared.

7. Step 5 – Generate Secondary Labels

The Computer Vision system may produce many labels for each image.

Example labels:

building
roof
tree
water
storm
sky
debris

These labels are called secondary labels because they are simply raw outputs from the vision system.

However, not all labels are equally important.

Example:

sky – not important for disaster analysis
tree – not very relevant
debris – very important indicator of damage

Therefore the system must determine which labels are important.

8. Step 6 – NLP Model Calculates Similarity

To determine which labels are important, the system uses Natural Language Processing (NLP).

It uses word embedding models such as:

- GloVe
- Word2Vec

These models understand relationships between words.

Example:

The following words are similar in meaning:

hurricane
cyclone
typhoon
storm

Even if the image label says “storm”, the NLP model can understand that it is related to hurricane-type disasters.

9. TF-IDF Score Calculation

The system also uses TF-IDF (Term Frequency – Inverse Document Frequency).

TF-IDF helps measure how important a word is in a document or image description.

Formula:

$$\text{TFIDF}(t,d) = \text{TF}(t,d) \times \log(N / \text{DF}_t)$$

Where:

$\text{TF}(t,d)$ = how often the word appears in the image labels

DF_t = number of images containing that word

N = total number of images

Practical idea

Common words → lower importance

Rare but meaningful words → higher importance

Example labels from an image:

building

roof

storm

water

debris

TF-IDF scores may be:

building – 4

storm – 9

roof – 11

debris – 19

This means debris is the most important indicator for this image.

10. Step 7 – Assign Primary Labels

After identifying important labels, the system assigns primary labels.

Primary labels represent the main categories of the image.

The paper defines five main categories.

Entity

People (E1)

Building (E2)

Vehicle (E3)

Hazard

Hurricane (H1)

Flood (H2)

Image Type

Ground (I1)

Aerial (I2)

Land Use

Residential (L1)

Commercial (L2)

Vegetation (L3)

Building Component

Roof (C1)

Wall (C2)

Window (C3)

Door (C4)

Garage (C5)

Example output for an image:

Entity: Building

Hazard: Hurricane

Image Type: Ground

Land Use: Residential

Component: Roof

Meaning:

Ground photo of a residential building with roof damage caused by a hurricane.

11. Step 8 – Store Results in JSON Database

After classification, the system stores the results in JSON format.

JSON stands for JavaScript Object Notation.

It is a structured way of storing data.

Example JSON output:

```
{  
  "entity": "building",  
  "hazard": "hurricane",  
  "image_type": "ground",  
  "land_use": "residential",  
  "component": "roof",  
  "tfidf_score": 19.26  
}
```

12. Why JSON Is Used

JSON is used because it is:

- simple
- easy for computers to read
- easy to store in databases
- easy to search

Example search query:

"hurricane + residential + roof"

The system checks the JSON metadata and returns matching images.

13. Real-World Example of the Full Process

Example image shows:

- damaged house
- broken roof
- debris
- flood water

The system processes it like this:

Step 1: Image uploaded

Step 2: Image stored in cloud

Step 3: AI detects labels (building, debris, flood)

Step 4: Faces or text blurred if detected

Step 5: Secondary labels generated

Step 6: NLP calculates similarity and TF-IDF

Step 7: Primary labels assigned

Step 8: Results saved as JSON metadata

Final classification:

Entity: Building

Hazard: Hurricane

Image Type: Ground

Land Use: Residential

Component: Roof

14. System Accuracy Validation

The researchers tested the system using 160 images.

Images were grouped into:

Affected – 40

Minor Damage – 40

Major Damage – 40

Destroyed – 40

Eight human experts manually labeled the images.

The system's results were compared with human labels.

Average difference:

13.32%

This means the automated system is very close to human accuracy.

15. Processing Cost and Speed

The system is efficient.

Processing performance:

1000 images

4 minutes processing time

\$2.80 cost

Manual labeling would take many hours or days, so automation is much faster.

16. Limitations

The system performs well for moderate damage but struggles with severely destroyed buildings.

Example results:

Affected –12.43% difference

Minor –10.24% difference

Major –16.88% difference

Destroyed –48.47% difference

This happens because extreme damage is difficult for AI to recognize accurately.

17. Applications of the System

This technology can help in:

- disaster damage assessment
- structural engineering research
- insurance claim analysis
- emergency response planning
- large-scale disaster image databases

18. Final Summary

The paper proposes an automated system that:

1. Receives disaster images
2. Uses computer vision to detect objects
3. Uses NLP and TF-IDF to identify important labels
4. Assigns structured disaster categories
5. Stores results in JSON format for easy search

This allows researchers to quickly find relevant disaster images without manually searching thousands of files.

FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding

After a flood disaster, many images are taken using drones (UAV).

But there was no proper dataset to train AI models to understand those images.

Purpose:

Use machine learning and computer vision to detect:

- flooded buildings
- flooded roads
- vehicles
- trees
- water areas

Example practical situation:

After a hurricane:

- thousands of drone photos captured
- humans cannot analyze all images quickly

AI can automatically detect flood damage.

Why FloodNet was Created

Before FloodNet, popular datasets existed like:

- ImageNet
- MS COCO
- PASCAL VOC
- Cityscapes

But :

Those datasets contain normal city images, not disaster scenes.

Example:

Cityscapes dataset contains:

- cars
- traffic lights
- roads
- buildings

But no flooded houses or flooded roads.

So they are not suitable for disaster damage assessment.

3. Problem with Existing Disaster Datasets

Older disaster datasets had problems.

Problem 1 – Satellite images

Satellite images are:

- low resolution
- objects look very small

Example:

A building may appear as few pixels only.

So it is difficult to detect:

- roof damage
- flooded building
- vehicles

Problem 2 – Social media images

Social media disaster images are messy.

Example:

Twitter / Facebook images:

- random angles
- poor quality
- no location information

- duplicate images

So researchers cannot easily use them for AI training.

Solution

Use UAV (drone) images.

Drone advantages:

- high resolution
- clear view
- closer to ground
- better damage visibility

So researchers created FloodNet dataset using drone images.

4. Data Collection (Very Important)

Images were collected after:

Hurricane Harvey (2017)

Location:

Texas, USA

Drone used:

DJI Mavic Pro

Drone flying height:

200 feet

This is much lower than satellite images.

So images are very clear and detailed.

5. Dataset Size

FloodNet contains:

3200 images

Each image resolution:

4000 × 3000 pixels

That is very high resolution.

Example:

You can clearly see:

- buildings
- trees
- vehicles
- water
- roads

6. Image Classes (Objects in Dataset)

Dataset contains 9 classes.

Classes mean types of objects AI must detect.

Background

Flooded building

Non-flooded building

Flooded road

Non-flooded road

Water

Tree

Vehicle

Grass

Pool

Example image may contain:

- flooded building
- water
- tree
- vehicle

AI must label each pixel.

7. Semantic Segmentation (Important Concept)

Semantic segmentation means:

AI classifies every pixel of the image.

Example:

Drone image:

building building building

water water water

tree tree

road road road

AI output becomes colored segmentation map.

Example:

- blue → water
- red → building
- green → tree
- grey → road

This helps accurately measure flooded areas.

8. Flooded Building Definition

Researchers defined a rule.

A building is flooded if:

At least one side of the building touches flood water.

Example:

Case 1:

Water touching building → Flooded

Case 2:

Water far from building → Not flooded

9. Image Classification Task

Another task is image classification.

Example:

Entire image is labeled as:

- Flooded scene
or
- Non flooded scene

Rule used:

If more than 30% of image area contains flood water → Flooded

10. Annotation Process

Annotation means manual labeling.

Researchers used a platform called:

V7 Darwin

Process:

Annotators label objects

Second person checks labels

Quality control done

So dataset becomes high quality.

11. Dataset Split

Dataset divided into:

70% → training data

30% → testing and validation

Purpose:

Training data → train AI model

Test data → evaluate model performance

12. Visual Question Answering (VQA)

FloodNet also supports Visual Question Answering.

Meaning:

AI answers questions based on images.

Example:

Drone image shows flooded houses.

Question:

"How many flooded buildings are in the image?"

AI must answer:

"3"

Another example:

Question:

"Is the road flooded?"

Answer:

"Yes"

13. Types of Questions

There are 3 question types.

1. Simple counting

Example:

How many buildings are there?

Answer: 5

2. Complex counting

Example:

How many flooded buildings are there?

Answer: 2

3. Condition recognition

Example questions:

Is the road flooded?

Is the image flooded?

Answers:

Yes / No

14. Dataset Question Size

FloodNet contains:

11,000 question-image pairs

Average:

3.5 questions per image

15. Algorithms Tested in Paper

Researchers tested multiple deep learning models.

For classification

Models used:

- ResNet50
- InceptionNetV3
- Xception

Best model:

ResNet50

Accuracy:

93.69%

For semantic segmentation

Models used:

- ENet
- DeepLabV3+
- PSPNet

Best model:

PSPNet

mIoU score:

79.69%

16. What is mIoU

mIoU means:

Mean Intersection over Union

It measures segmentation accuracy.

Formula idea:

Overlap area / Total combined area

Example:

Real flooded area = 100 pixels

Model predicted flood = 90 pixels

Overlap = 80 pixels

IoU:

$80 / 110$

Higher IoU $\rightarrow$ better model.

17. Visual Question Answering Models

Models tested:

- Feature concatenation
- Element wise multiplication

- Stacked Attention Network
- MFB with Co-Attention

Best model:

MFB with Co-Attention

Accuracy:

73%

18. Challenges in FloodNet

Researchers found some difficulties.

Small objects

Example:

- vehicles
- swimming pools

These objects are very small in aerial images.

So AI struggles to detect them.

Flood detection difficulty

From top view drone image, it is difficult to see:

- whether building is flooded
- whether road is flooded

Because sometimes water reflections look similar.

19. Contribution of FloodNet

FloodNet is important because it is:

- ✓ High resolution UAV dataset
- ✓ Post-disaster imagery
- ✓ Supports 3 tasks:

- classification
- semantic segmentation
- visual question answering

This was first dataset combining all three tasks.

20. Why FloodNet is Important for Your FYP

If your project is Post-Disaster Damage Assessment, this dataset helps in:

- flood detection
- building damage assessment
- road flooding analysis
- disaster response planning

Example use cases:

AI can automatically detect:

- flooded houses
- blocked roads
- vehicles in flood zones

This helps rescue teams make faster decisions.